Example 4.25 Evaluate:

$$\text{(i) } {}^4P_4 \quad \text{(ii) } {}^5P_3 \quad \text{(iii)} {}^8P_4 \quad \text{(iv) } {}^6P_5.$$

Solution:

(i) ${}^4P_4 = 4 \times 3 \times 2 \times 1 = 4! = 24.$

(ii) ${}^5P_3 = 5 \times 4 \times 3 = 60.$

(iii) ${}^8P_4 = 8 \times 7 \times 6 \times 5 = 1680.$

(iv) ${}^6P_5 = 6 \times 5 \times 4 \times 3 \times 2 = 6! = 720.$

Example 4.26 If ${}^{(n+2)}P_4 = 42 \times {}^nP_2$, find n.

Solution:

$$\begin{aligned} {}^{(n+2)}P_4 &= 42 \times {}^nP_2 \\ \Rightarrow \frac{{}^{(n+2)}P_4}{{}^nP_2} &= 42 \\ \Rightarrow \frac{(n+2)(n+1)(n)(n-1)}{n(n-1)} &= 42 \\ \Rightarrow (n+2)(n+1) &= 42 = 7 \times 6 \\ \Rightarrow n+2 &= 7 \Rightarrow n = 5. \end{aligned}$$

Example 4.27 If ${}^{10}P_r = {}^7P_{r+2}$ find r.

Solution:

$$\begin{aligned} {}^{10}P_r &= {}^7P_{r+2} \\ \frac{10!}{(10-r)!} &= \frac{7!}{(5-r)!} \end{aligned}$$

i.e.,

$$\frac{10 \times 9 \times 8 \times 7!}{(10-r) \times (9-r) \times (8-r) \times (7-r) \times (6-r) \times (5-r)!} = \frac{7!}{(5-r)!}$$

$$(10-r) \times (9-r) \times (8-r) \times (7-r) \times (6-r) = 10 \times 9 \times 8 = 6 \times 5 \times 4 \times 3 \times 2.$$

Therefore, $10 - r = 6 \Rightarrow r = 4$.

Example 4.28 How many 'letter strings' together can be formed with the letters of the word "VOWELS" so that

(i) the strings begin with E

(ii) the strings begin with E and end with W.